MAHATHMA GANDHI UNIVERSITY KOTTAYAM

MGU – UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3DSEECT200 Analog Electronics

PRACTICAL EXAMINATION -MODEL QUESTION PAPER

Duration: 2 Hours **Maximum Marks: 35**

INSRUCTIONS

- Demonstrate the given task (Any of the tasks listed below)
- Viva Section (Minimum 5 questions from the practical module)
- Certified Lab report

EVALUATION CRITERIA

- **Viva:** 7 Marks
- **Lab report:** 8 Marks
- **Demonstration:** 20 Marks.

EXPERIMENTS: (Choose any one)

1. Construct an RC differentiator circuit and verify its input and output waveforms.
2. Construct an RC integrator circuit and verify its input and output waveforms.
3. Design and set up a low-pass filter using an RC circuit with suitable cutoff frequencies.
4. Design and set up a high-pass filter using an RC circuit with suitable cut-off frequencies.
5. Design an Inverting amplifier using an op amp and find its bandwidth.
6. Design a non-inverting amplifier using an op amp and find its bandwidth.
7. Design and set up a voltage comparator using an op-amp.
8. Design and set up an Astable multivibrator using IC 555.
9. Design and set up a Schmitt trigger with suitable threshold points using an op amp.
10. Design and set up a squarewave generator of suitable frequency